



KRISHIBOARD

Farm Produce Management System

Product Requirements Document

Version 1.0

April 2026

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Document Status	Draft
Platform	ServiceNow (UiPage React Portal)
Target Users	Farm Owners, Agricultural Officers
Deployment	ServiceNow SDK + Fluent API

1. Executive Summary

KrishiBoard is a single-page React portal application built on the ServiceNow platform, designed for small-to-medium farm owners to manage their produce, track budgets, forecast yields, and plan seasonal crop rotations. The application features an interactive farm map divided into 7 configurable plots, supporting 2 grain crops (Wheat, Rice), 1 oilseed crop (Mustard), and 2 vegetable gardens (Tomato, Spinach).

The system provides real-time visibility into cultivated area, seed and fertilizer requirements, budget breakdowns, expected vs. actual yield tracking, monthly income/expense forecasting, and a seasonal crop calendar. All data is persisted in ServiceNow custom tables and accessed via the Table REST API, making it enterprise-grade, auditable, and extensible.

2. Product Vision

2.1 Problem Statement

Indian farmers managing 5–15 acre holdings typically track crop data in paper notebooks or fragmented spreadsheets. There is no unified system that combines spatial plot management, financial forecasting, and seasonal planning into a single intuitive interface. This results in over-purchasing of seeds/fertilizers, missed planting windows, and inaccurate revenue projections.

2.2 Solution

KrishiBoard delivers a modern, card-based dashboard (inspired by Berry Dashboard) as a ServiceNow UiPage React portal. It provides visual farm plot management, automated input calculations, financial forecasting, and a seasonal planner — all within the ServiceNow ecosystem for enterprise-grade security, workflow automation, and audit trails.

2.3 Key Value Propositions

- Interactive SVG farm map with 7 clickable, assignable plots showing crop distribution at a glance
- Automated seed, fertilizer, and labor cost calculations based on crop type and plot area
- Expected vs. current yield tracking with visual progress bars per plot
- Monthly income/expense forecast charts for full-season financial planning
- Seasonal crop calendar showing optimal planting windows for all 5 supported crops
- Profitability analysis per crop with margin percentages and ROI calculations
- Seamless single-page experience with no page reloads, smooth transitions, and Berry-inspired warm UI

3. User Personas

Persona	Role	Goals	Pain Points
Rajesh (45)	Farm Owner 10 acres, Haryana	Track all plots in one place; know exact seed/fert needs; forecast income for loan planning	Paper records lost; buys excess fertilizer; no visibility into profit margins
Priya (32)	Agricultural Officer Block Development Office	Monitor multiple farm dashboards; generate seasonal reports; advise on crop rotation	Manual data collection; inconsistent farmer records; no regional aggregation
Vikram (28)	Farm Manager Contract farming	Assign crops to plots; track daily yields; manage budgets across multiple holdings	Spreadsheet chaos; missed seasonal windows; no real-time yield tracking

4. Feature Requirements

4.1 Interactive Farm Map

The centerpiece of KrishiBoard is an SVG-based overhead farm map divided into 7 distinct plots. Each plot is rendered as a clickable polygon with color-coding based on the assigned crop, emoji indicators, and area labels. The map includes a compass indicator, river feature, and textured fill patterns for planted plots.

Feature	Description	Priority
Plot Click	Clicking a plot opens a detail panel showing crop info, yield progress, revenue estimate, and seed/fertilizer requirements	P0
Crop Assignment	Empty (fallow) plots show an Assign Crop button that opens a modal with all 5 crop options and their key stats	P0
Crop Removal	Active plots can have their crop removed, resetting to fallow status	P1
Hover Effects	Plots highlight on hover with increased fill opacity and border color change	P1
Legend	Color-coded legend below the map showing all crop types and fallow land	P0
Area Labels	Each plot displays its name, crop emoji, and acreage directly on the map	P0

4.2 Dashboard KPI Cards

Four top-level KPI cards provide instant visibility into key farm metrics. Each card features an icon with colored background, primary value, subtitle, and optional trend indicator.

KPI Card	Metric	Calculation	Example
Cultivated Area	Acres under	Sum of area for plots with	7.5 ac of 10 acres (75%)

	cultivation	assigned crops	
Active Crops	Distinct crop varieties	Count of unique crops across all plots	5 varieties
Season Budget	Total estimated cost	Sum of (seeds + fertilizer + labor) per plot	₹1.2L
Expected Revenue	Projected income	Sum of (expectedYield × area × MSP) per crop	₹5.8L (384% ROI)

4.3 Budget and Forecasting

The budget module provides both aggregate and per-crop financial analysis:

1. Budget Breakdown Pie Chart: Visualizes total spend split across Seeds, Fertilizer, and Labor categories
2. Monthly Forecast Area Chart: Projects income vs. expense month-by-month across the full season based on expected harvest months and evenly distributed costs
3. Per-Crop Profitability Cards: Individual cards per active crop showing profit amount and margin percentage with color coding (green for profitable, amber for low margin)
4. Revenue estimates use current Minimum Support Price (MSP) rates from the Government of India

4.4 Seed and Fertilizer Requirements

A comprehensive table listing all active plots with their complete input requirements. This serves as the farmer's purchase list for the season.

Column	Description	Source
Plot Name	Name and plot number	x_krishi_plot.name
Crop	Assigned crop with type badge	x_krishi_crop_master.name
Area	Plot area in acres	x_krishi_plot.area_acres
Seeds Needed	seedRate × area	Calculated from crop master
Fertilizer Plan	Specific fertilizer combination	x_krishi_crop_master.fertilizer_plan
Fertilizer Cost	fertCost × area	Calculated from crop master
Water Requirement	Low / Medium / High	x_krishi_crop_master.water_need
Estimated Revenue	expectedYield × area × MSP	Calculated

4.5 Seasonal Crop Calendar

A 12-month grid showing optimal planting and growing windows for each crop. Active months are highlighted with the crop's color and a checkmark indicator. This helps farmers plan crop rotation and avoid planting outside optimal windows.

4.6 Yield Tracking

A grouped bar chart comparing Expected yield (pale green) vs. Current yield (dark green) in quintals for each active crop. The plot detail panel also shows a progress bar with percentage completion toward the expected yield target.

5. Data Model — ServiceNow Table Design

The application requires 5 custom tables in ServiceNow, all scoped under x_krishi_farm. Below are the complete table schemas with field types, constraints, and relationships.

5.1 Farm Profile (x_krishi_farm_profile)

Stores the farm-level configuration and owner information. One record per farm.

Field Name	Label	Type	Max Length	Mandatory	Default/Notes
x_krishi_farm_name	Farm Name	String	200	Yes	—
x_krishi_farm_owner	Owner	Reference	—	Yes	Ref: sys_user
x_krishi_farm_total_area	Total Area (acres)	Decimal	—	Yes	Sum of all plot areas
x_krishi_farm_location	Location	String	500	No	Village, District, State
x_krishi_farm_geo_lat	Latitude	Decimal	—	No	GPS coordinates
x_krishi_farm_geo_long	Longitude	Decimal	—	No	GPS coordinates
x_krishi_farm_soil_type	Soil Type	Choice	—	No	Alluvial / Black / Red / Laterite
x_krishi_farm_water_source	Water Source	Choice	—	No	Canal / Borewell / Rainfed / River
x_krishi_farm_season	Current Season	String	100	No	e.g., Rabi 2025-26
x_krishi_farm_notes	Notes	Journal	—	No	General farm notes

5.2 Plot Master (x_krishi_plot)

Stores individual plot/field information. Each farm can have multiple plots (7 in our demo configuration).

Field Name	Label	Type	Max Length	Mandatory	Default/Notes
x_krishi_plot_name	Plot Name	String	200	Yes	e.g., North Field
x_krishi_plot_farm	Farm	Reference	—	Yes	Ref: x_krishi_farm_profile
x_krishi_plot_area	Area (acres)	Decimal	—	Yes	—
x_krishi_plot_crop	Current Crop	Reference	—	No	Ref: x_krishi_crop_master (null = fallow)
x_krishi_plot_status	Status	Choice	—	Yes	fallow / growing / flowering / harvesting /

					resting
x_krishi_plot_current_yield	Current Yield (qt)	Decimal	—	No	Updated periodically; 0 if fallow
x_krishi_plot_planted_date	Planted Date	Date	—	No	When crop was planted
x_krishi_plot_expected_harvest	Expected Harvest	Date	—	No	Projected harvest date
x_krishi_plot_svg_path	SVG Path Data	String	1000	No	SVG polygon path for map rendering
x_krishi_plot_label_x	Label X Position	Integer	—	No	SVG label X coordinate
x_krishi_plot_label_y	Label Y Position	Integer	—	No	SVG label Y coordinate
x_krishi_plot_sort_order	Display Order	Integer	—	No	Ordering on map
x_krishi_plot_notes	Notes	Journal	—	No	Plot-specific notes

5.3 Crop Master (x_krishi_crop_master)

Reference table of all supported crops with their agricultural parameters, cost data, and seasonal windows. This is the core configuration table that drives all calculations.

Field Name	Label	Type	Mandatory	Default/Notes
x_krishi_crop_name	Crop Name	String	Yes	Wheat, Rice, Mustard, Tomato, Spinach
x_krishi_crop_type	Crop Type	Choice	Yes	Grain / Oilseed / Vegetable / Pulse / Fruit
x_krishi_crop_emoji	Display Emoji	String	No	Visual indicator for UI
x_krishi_crop_color	UI Color Code	String	No	Hex color for map/charts (e.g., #E9C46A)
x_krishi_crop_season	Growing Season	String	Yes	e.g., Rabi (Oct–Mar)
x_krishi_crop_seed_rate	Seed Rate (per acre)	Decimal	Yes	kg per acre
x_krishi_crop_seed_unit	Seed Unit	String	No	kg/acre
x_krishi_crop_fert_plan	Fertilizer Plan	String	Yes	e.g., DAP 50kg + Urea 60kg
x_krishi_crop_fert_cost	Fertilizer Cost/acre	Decimal	Yes	INR per acre
x_krishi_crop_expected_yield	Expected Yield/acre	Decimal	Yes	Quintals per acre
x_krishi_crop_msp	MSP (per quintal)	Decimal	Yes	Government MSP rate in INR
x_krishi_crop_water_need	Water Requirement	Choice	Yes	Low / Medium / High

x_krishi_crop_months_active	Active Months	String	No	Comma-separated: 0,0,1,0,0,0,0,0,1,1,1
x_krishi_crop_labor_cost	Labor Cost/acre	Decimal	No	INR per acre per season; default 4500
x_krishi_crop_seed_price_kg	Seed Price/kg	Decimal	No	INR per kg; default 45

5.4 Season Record (x_krishi_season_record)

Tracks actual financial and yield data per plot per season. This is the transactional table that captures real outcomes for historical analysis and forecasting.

Field Name	Label	Type	Mandatory	Notes
x_krishi_sr_plot	Plot	Reference	Yes	Ref: x_krishi_plot
x_krishi_sr_crop	Crop Planted	Reference	Yes	Ref: x_krishi_crop_master
x_krishi_sr_season	Season	String	Yes	e.g., Rabi 2025-26
x_krishi_sr_area	Area Planted (ac)	Decimal	Yes	Snapshot of plot area at planting
x_krishi_sr_planted_date	Planted Date	Date	Yes	—
x_krishi_sr_harvest_date	Harvest Date	Date	No	Filled after harvest
x_krishi_sr_expected_yield	Expected Yield (qt)	Decimal	Yes	From crop master at time of planting
x_krishi_sr_actual_yield	Actual Yield (qt)	Decimal	No	Filled after harvest
x_krishi_sr_seed_cost	Seed Cost (INR)	Decimal	Yes	Calculated: rate × area × price/kg
x_krishi_sr_fert_cost	Fertilizer Cost	Decimal	Yes	Calculated: fertCost × area
x_krishi_sr_labor_cost	Labor Cost	Decimal	Yes	Calculated: laborRate × area
x_krishi_sr_other_cost	Other Expenses	Decimal	No	Irrigation, pesticides, transport
x_krishi_sr_total_cost	Total Cost	Decimal	Yes	Sum of all costs
x_krishi_sr_revenue	Actual Revenue	Decimal	No	actualYield × selling price
x_krishi_sr_profit	Profit/Loss	Decimal	No	Revenue minus Total Cost
x_krishi_sr_status	Status	Choice	Yes	planned / active / harvested / cancelled

5.5 Budget Entry (x_krishi_budget)

Itemized budget entries for granular expense tracking. Each season record can have multiple budget line items for detailed financial management.

Field Name	Label	Type	Mandatory	Notes
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x_krishi_bgt_season_record	Season Record	Reference	Yes	Ref: x_krishi_season_record
x_krishi_bgt_category	Category	Choice	Yes	Seeds / Fertilizer / Labor / Pesticide / Irrigation / Transport / Equipment / Other
x_krishi_bgt_description	Description	String	Yes	e.g., DAP 50kg bags x 2
x_krishi_bgt_estimated_amt	Estimated Amount	Decimal	Yes	Budgeted INR
x_krishi_bgt_actual_amt	Actual Amount	Decimal	No	Actual spend INR
x_krishi_bgt_vendor	Vendor/Supplier	String	No	Supplier name
x_krishi_bgt_purchase_date	Purchase Date	Date	No	—
x_krishi_bgt_receipt	Receipt/Invoice	Attachment	No	Uploaded image/PDF
x_krishi_bgt_notes	Notes	String	No	—

6. Demo Data

Below is the complete demo dataset used to populate the application for demonstration and testing purposes. All prices reflect 2025-26 MSP rates from the Government of India.

6.1 Crop Master — Demo Records

Crop	Type	Season	Seed Rate	Fert. Plan	Fert. Cost/ac	Yield /ac	MSP/qt	Water
Wheat	Grain	Rabi (Oct–Mar)	100 kg/ac	DAP 50kg + Urea 60kg	₹3,200	18 qt	₹2,275	Low
Rice	Grain	Kharif (Jun–Oct)	20 kg/ac	DAP 40kg + Urea 80kg	₹3,600	25 qt	₹2,183	High
Mustard	Oilseed	Rabi (Oct–Feb)	5 kg/ac	DAP 40kg + Urea 40kg	₹2,400	8 qt	₹5,650	Low
Tomato	Vegetable	Multi-season	0.5 kg/ac	NPK 100kg + Compost	₹8,500	120 qt	₹1,800	Medium
Spinach	Vegetable	Rabi (Oct–Mar)	15 kg/ac	Urea 40kg + Compost	₹2,800	60 qt	₹2,200	Medium

6.2 Plot Configuration — Demo Records

Plot #	Name	Area (ac)	Assigned Crop	Status	Current Yield	Yield %
1	North Field	2.0	Wheat	Growing	15.2 qt/ac	84%
2	East Paddy	2.5	Rice	Growing	21.0 qt/ac	84%
3	West Strip	1.5	Mustard	Flowering	6.5 qt/ac	81%
4	Kitchen Garden A	0.8	Tomato	Harvesting	95.0 qt/ac	79%
5	Kitchen Garden B	0.7	Spinach	Growing	48.0 qt/ac	80%
6	South Reserve	1.0	— (Fallow)	Fallow	—	—
7	Riverbank Plot	1.5	— (Fallow)	Fallow	—	—

Total Farm Area: 10.0 acres | **Cultivated:** 7.5 acres (75%) | **Fallow:** 2.5 acres (25%)

6.3 Budget Summary — Demo Data

Plot	Crop	Seed Cost	Fert. Cost	Labor Cost	Total Cost	Expected Revenue	Projected Profit	Margin
North Field	Wheat	₹9,000	₹6,400	₹9,000	₹24,400	₹81,900	₹57,500	70%
East Paddy	Rice	₹2,250	₹9,000	₹11,250	₹22,500	₹1,36,438	₹1,13,938	84%

West Strip	Mustard	₹338	₹3,600	₹6,750	₹10,688	₹67,800	₹57,112	84%
Kitchen Gdn A	Tomato	₹18	₹6,800	₹3,600	₹10,418	₹1,72,800	₹1,62,382	94%
Kitchen Gdn B	Spinach	₹473	₹1,960	₹3,150	₹5,583	₹92,400	₹86,817	94%
TOTALS	—	₹12,079	₹27,760	₹33,750	₹73,589	₹5,51,338	₹4,77,749	87%

6.4 Seasonal Calendar — Demo Data

The seasonal calendar shows active growing months for each crop. A value of 1 indicates the crop can be planted/is actively growing in that month; 0 indicates off-season.

Crop	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Wheat	✓	✓	✓	—	—	—	—	—	—	✓	✓	✓
Rice	—	—	—	—	—	✓	✓	✓	✓	✓	—	—
Mustard	—	✓	—	—	—	—	—	—	—	✓	✓	✓
Tomato	✓	✓	✓	—	—	—	✓	✓	✓	✓	✓	✓
Spinach	✓	✓	✓	—	—	—	—	—	—	✓	✓	✓

7. Technical Architecture

7.1 Technology Stack

Layer	Technology	Purpose
Frontend Framework	React 18+ (JSX)	Component-based SPA rendering
Charts	Recharts	PieChart, BarChart, AreaChart components
Icons	Lucide React	Lightweight SVG icon library
Styling	Inline CSS-in-JS + CSS Variables	Berry-inspired warm earth tone palette
Fonts	Fraunces (display) + DM Sans (body)	Google Fonts loaded dynamically
Map Rendering	Native SVG	Custom polygon paths for each plot
Platform	ServiceNow UiPage	Hosts the React SPA via SDK
API Layer	ServiceNow Table REST API	CRUD operations on custom tables
Auth	window.g_ck token	ServiceNow session-based authentication
Build Tool	ServiceNow SDK (now-sdk)	Fluent DSL compilation and deployment
State Management	React useState/useMemo	Local state with REST API persistence

7.2 API Endpoints

All data access uses the ServiceNow Table REST API at `/api/now/table/{tableName}` with the `X-UserToken` header (`window.g_ck`) for authentication. Key operations:

Operation	Method	Endpoint	Use Case
List Plots	GET	<code>/api/now/table/x_krishi_plot?sysparm_display_value=all</code>	Load farm map on dashboard init
Get Crop Master	GET	<code>/api/now/table/x_krishi_crop_master</code>	Populate crop selector modal
Assign Crop	PATCH	<code>/api/now/table/x_krishi_plot/{sys_id}</code>	Update plot with crop reference
Update Yield	PATCH	<code>/api/now/table/x_krishi_plot/{sys_id}</code>	Record current yield value
Create Season Record	POST	<code>/api/now/table/x_krishi_season_record</code>	Log new season entry
Add Budget Entry	POST	<code>/api/now/table/x_krishi_budget</code>	Track expense line item
Get Season History	GET	<code>/api/now/table/x_krishi_season_record?plot={id}</code>	Historical yield/cost data

7.3 ServiceNow SDK Project Structure

The application follows the standard ServiceNow SDK scoped app structure with UiPage for React hosting:

- src/fluent/tables/ — Table definitions for all 5 custom tables (farm_profile, plot, crop_master, season_record, budget)
- src/fluent/acls/ — ACL rules restricting table access to farm_user and admin roles
- src/fluent/records/ — Seed data records for crop master and demo plot configuration
- src/fluent/ui-pages/ — UiPage definition referencing the React entry point
- src/client/ — React application code (index.html, main.tsx, App.tsx, components/)
- assets/ — Farm logo SVG and any static images

8. UI/UX Design Specifications

8.1 Design System

The design is inspired by Berry Dashboard (by CodedThemes) with an agricultural twist. The palette uses earthy greens, warm ambers, and terracotta accents on a cream background, evoking natural farmland aesthetics while maintaining professional dashboard readability.

Token	Value	Usage
Primary	#2D6A4F	Sidebar, headings, active states, profit indicators
Primary Light	#40916C	Hover states, progress bars, trend arrows
Primary Pale	#D8F3DC	Badge backgrounds, yield bar track, revenue cards
Amber	#E76F51	Warnings, harvest status, remove actions
Amber Light	#F4A261	Secondary accents, expense charts, flowering status
Sky	#264653	Sidebar gradient end, subheadings
Sky Light	#2A9D8F	Teal accents, KPI card backgrounds
Brown	#6D4C41	Fallow plot indicators, earth tones
Background	#F8F6F1	Page background (warm cream)
Card	#FFFFFF	Card surfaces
Card Alt	#F1EDE4	Detail panel sub-cards, alternating rows
Border	#E8E2D6	Card borders, table dividers
Text	#1B2420	Primary body text
Text Mid	#4A5859	Labels, secondary text
Text Light	#8A9A9B	Captions, placeholder text

8.2 Typography

Element	Font	Weight	Size
App Brand (KrishiBoard)	Fraunces	700	22px
Page Titles	Fraunces	700	26px
Stat Values	Fraunces	700	28px

Card Titles	DM Sans	600	13px (uppercase, spaced)
Body Text	DM Sans	400	14px
Labels/Captions	DM Sans	400	11–12px
Badges	DM Sans	600	11px

8.3 Component Specifications

Component	Border Radius	Shadow	Padding
Cards	16px	0 2px 16px rgba(45,106,79,0.08)	24px
Badges	20px	None	3px 10px
Buttons (Primary)	10px	0 4px 14px rgba(45,106,79,0.3)	10px 20px
Crop Tags	20px	None	6px 14px
Modal	16px	0 24px 80px rgba(0,0,0,0.25)	32px
Icon Containers	14px	None	Centered 52×52px
Sub-cards (Detail Panel)	10px	None	14px

9. Success Metrics

Metric	Target	Measurement
Page Load Time	< 2 seconds	Lighthouse audit on ServiceNow instance
User Adoption	80% weekly active use	Session tracking via ServiceNow analytics
Data Accuracy	> 95% yield forecast accuracy	Compare expected vs. actual at harvest
Budget Variance	< 15% cost deviation	Actual spend vs. estimated budget
Plot Utilization	> 85% cultivated area	Fallow area / total area ratio
User Satisfaction	> 4.2/5 rating	In-app feedback survey

10. Future Roadmap

10.1 Phase 2 Features

1. Weather API integration (OpenWeather/IMD) for real-time weather overlays on the farm map
2. Soil health monitoring with IoT sensor data from connected soil probes
3. Multi-farm support for agricultural officers managing a portfolio of farms
4. Photo capture via mobile for crop health monitoring with AI-based disease detection
5. Market price ticker showing live mandi prices alongside MSP rates
6. Automated procurement workflow: generate purchase orders for seeds/fertilizer from the requirements table

7. Historical season comparison: side-by-side analytics across previous Rabi/Kharif seasons
8. Export to PDF: downloadable season report for bank loan documentation
9. Push notifications for planting windows, weather alerts, and harvest reminders
10. Multi-language support (Hindi, Marathi, Punjabi, Telugu) for broader farmer adoption

11. Glossary

Term	Definition
MSP	Minimum Support Price — government-guaranteed price for agricultural produce
Rabi	Winter cropping season (October to March) — wheat, mustard, spinach
Kharif	Monsoon cropping season (June to October) — rice, maize, cotton
Quintal (qt)	Unit of weight = 100 kg, commonly used for crop yields in India
DAP	Di-Ammonium Phosphate — common phosphorus-based fertilizer
NPK	Nitrogen-Phosphorus-Potassium — balanced compound fertilizer
Fallow	Agricultural land left unplanted for a season to restore soil fertility
UiPage	ServiceNow construct for hosting custom HTML/React pages in the platform
Fluent API	ServiceNow SDK's TypeScript DSL for defining metadata as code
g_ck	ServiceNow's client-side CSRF token used for REST API authentication

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